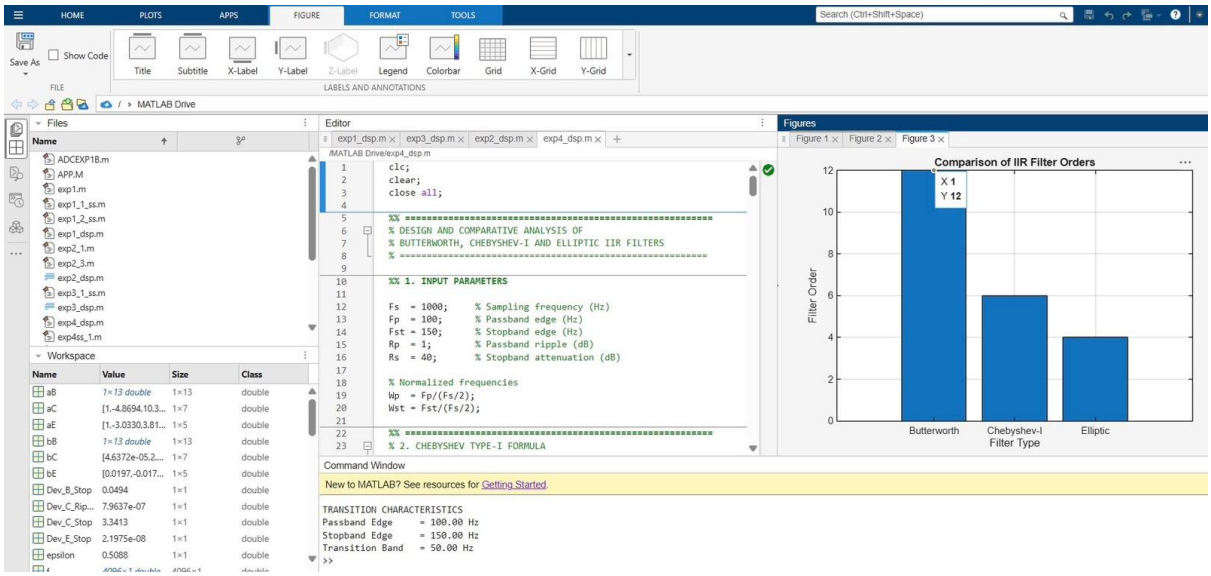
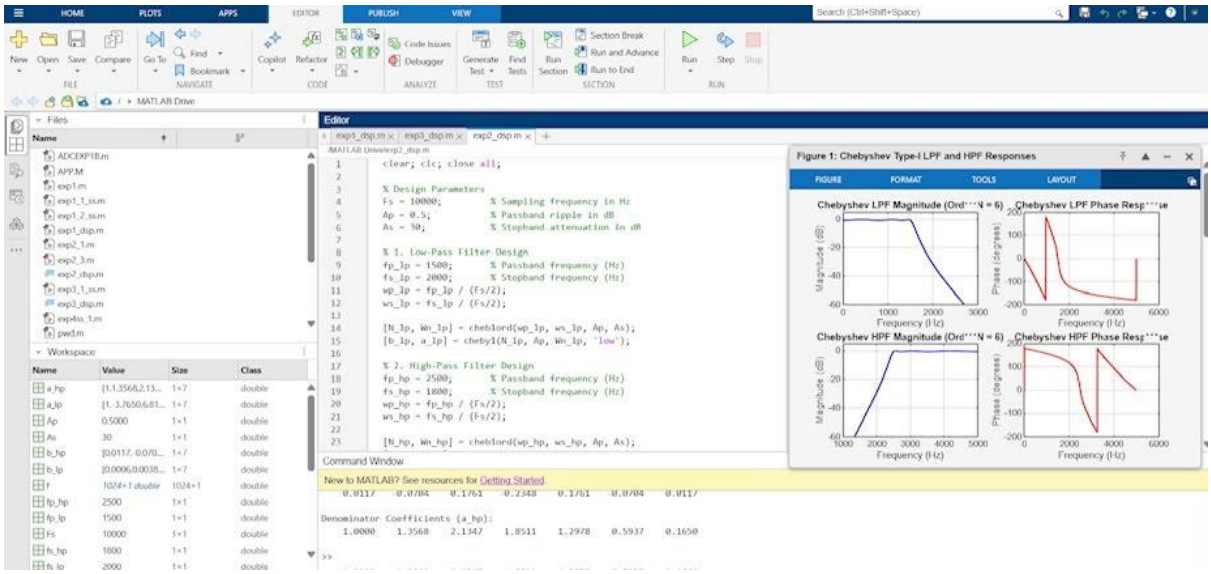
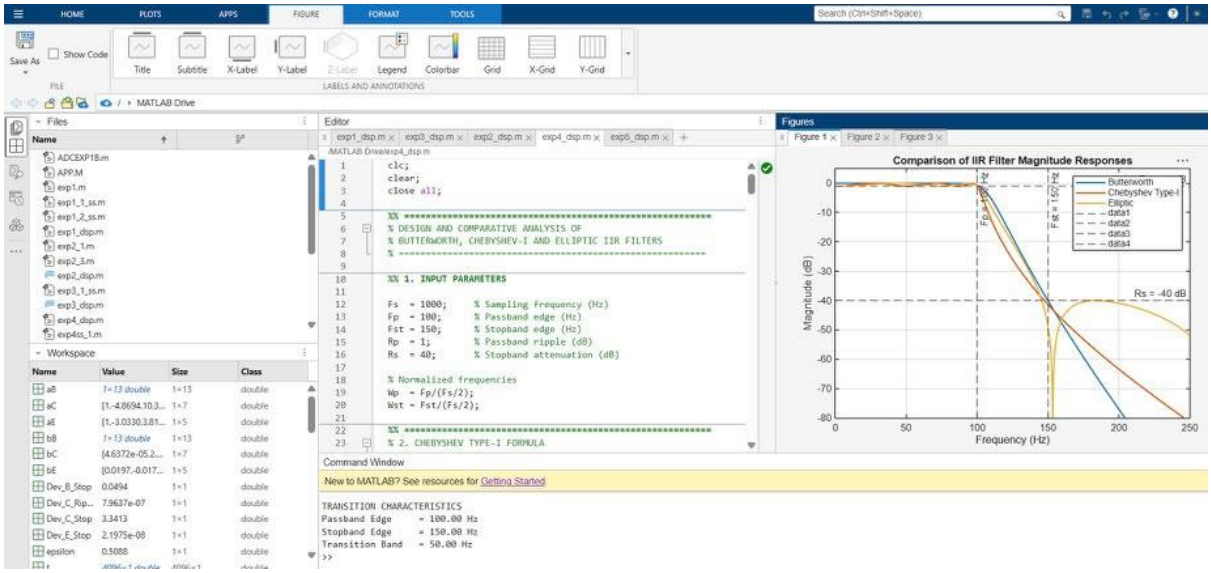
ECA0907 DIGITAL SIGNAL PROCESSING

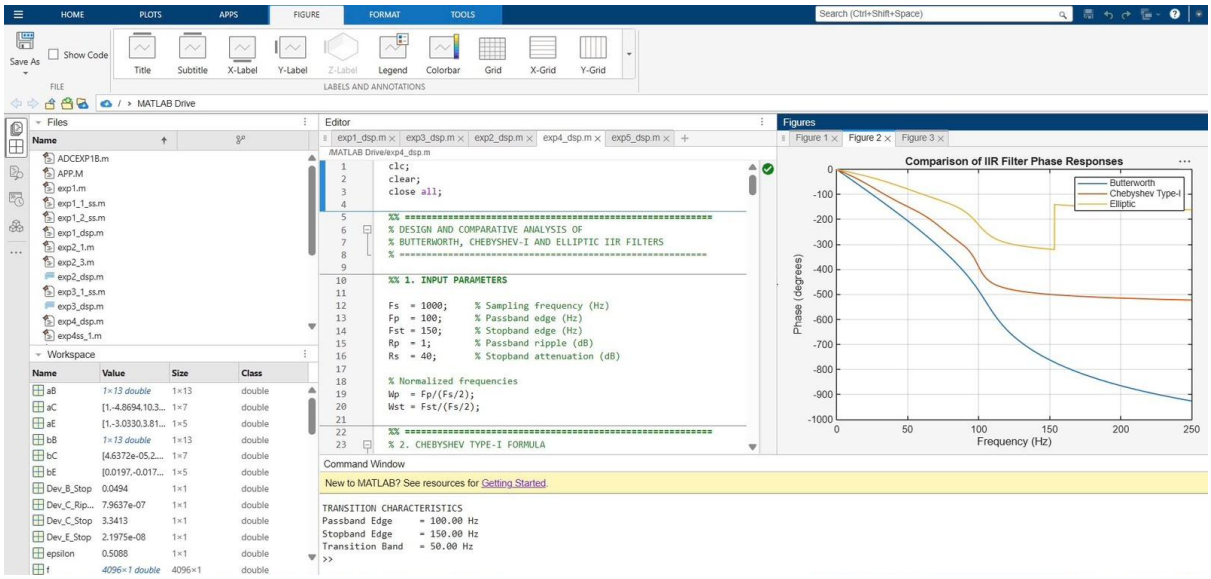
CO1 – AT2

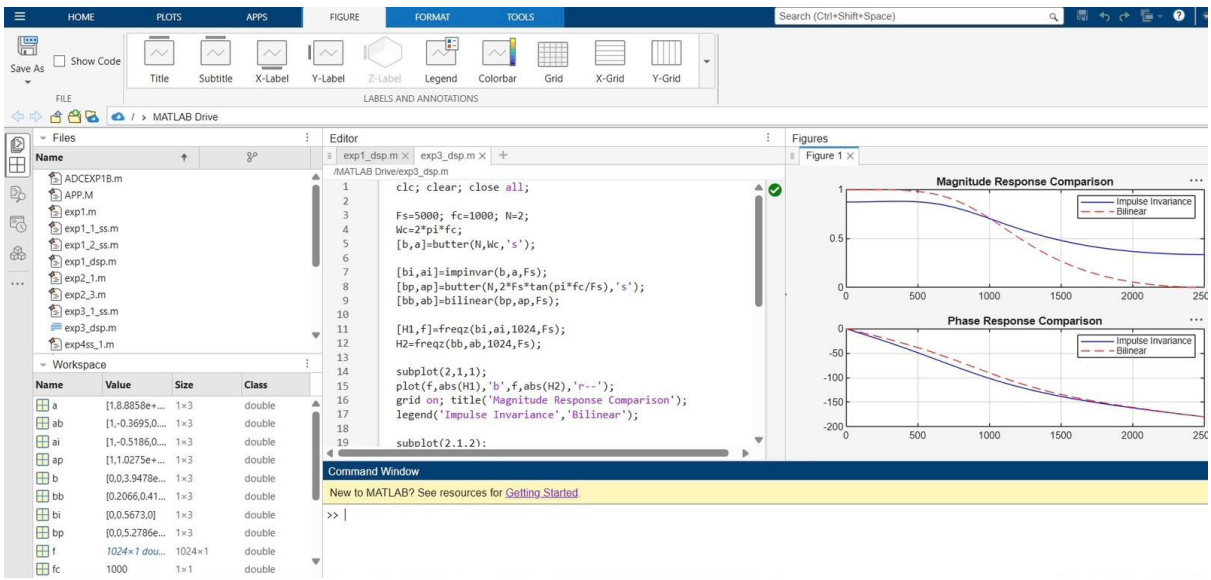
N. janardhan Reddy – 192312525

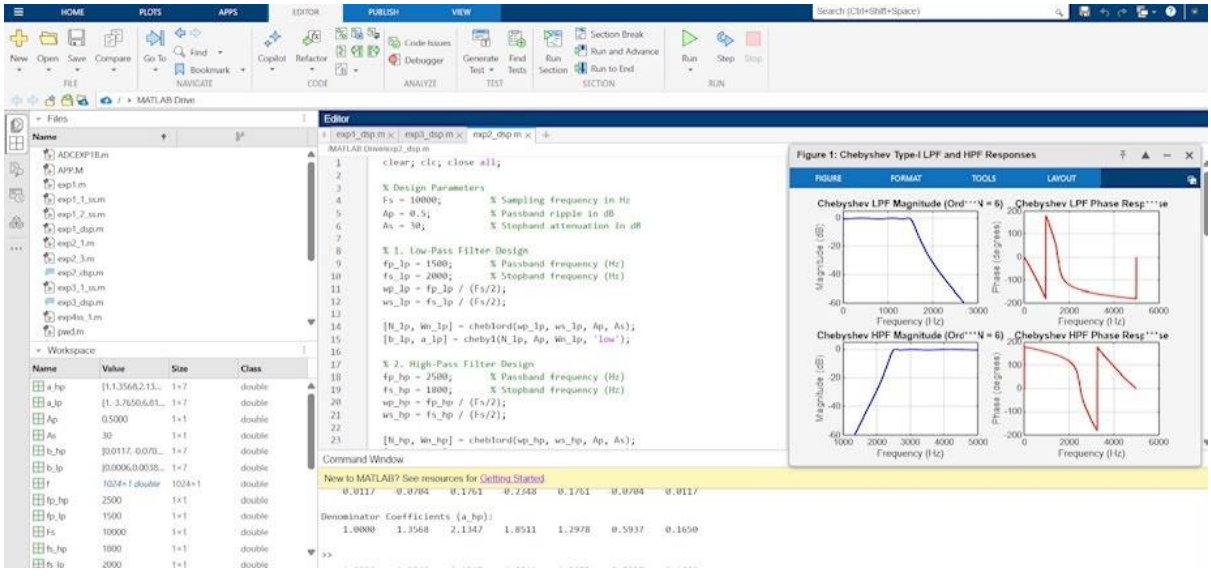


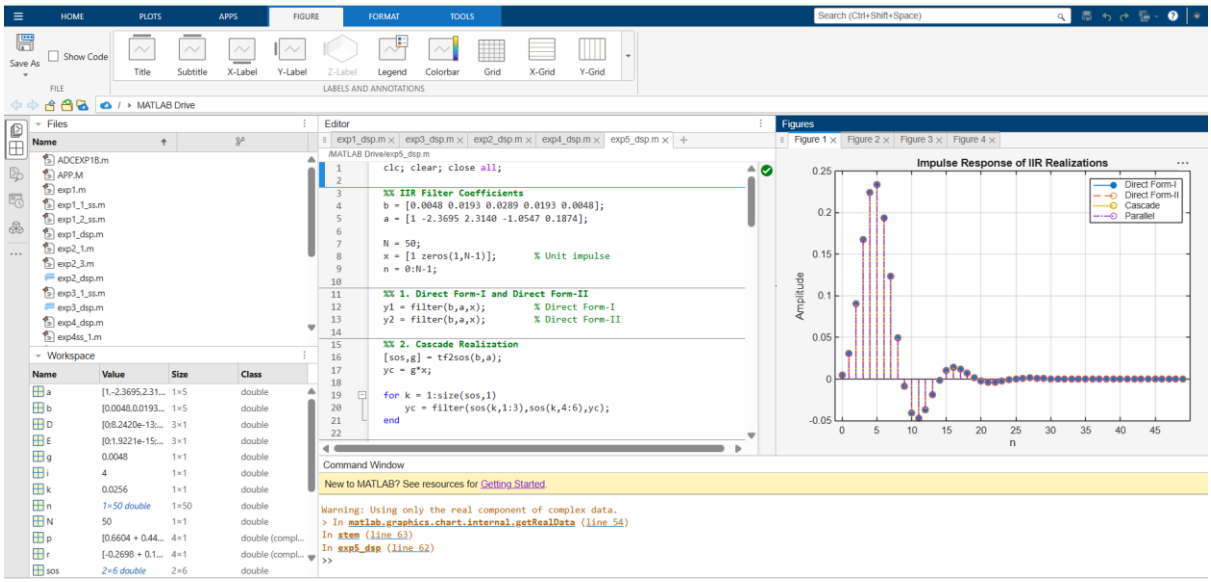


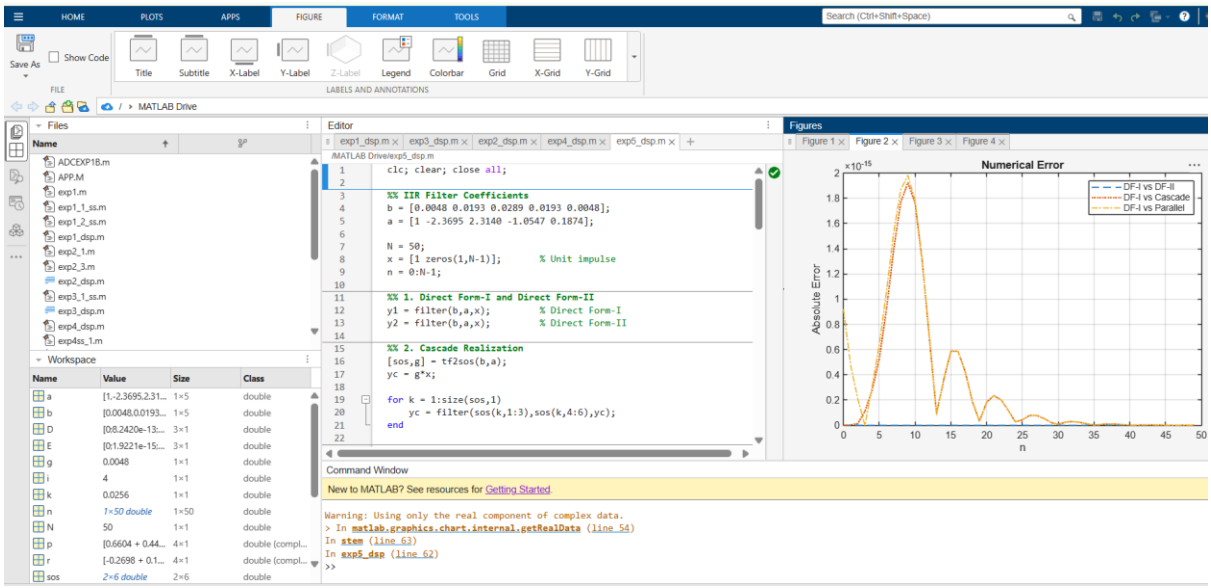

Comparison of IIR Filter Magnitude Responses
Butterworth
Chebyshev Type-I
Elliptic
Rs = 40 dB
Magnitude (dB)
Frequency (Hz)

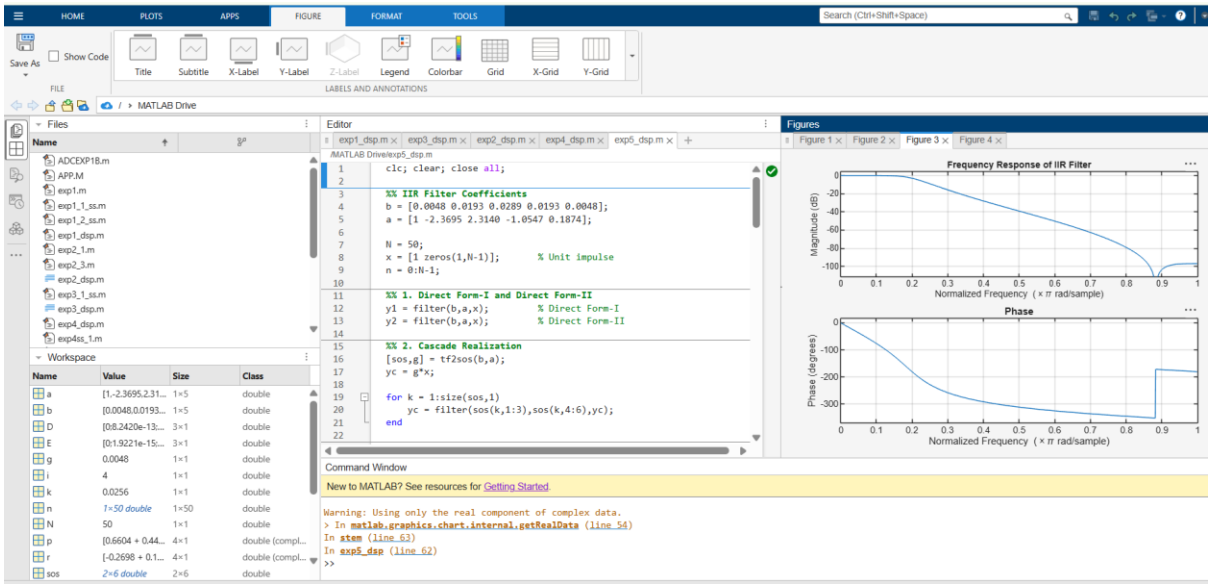

Comparison of IIR Filter Phase Responses
Butterworth
Chebyshev Type-I
Elliptic
Phase (degrees)
Frequency (Hz)

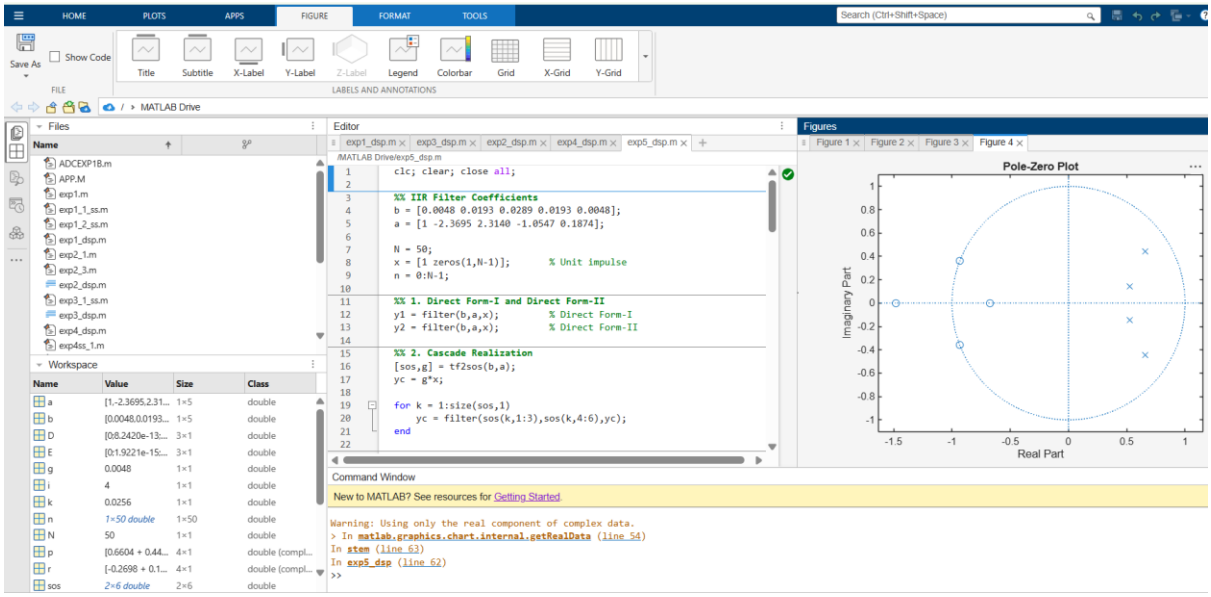

Magnitude Response Comparison
Impulse Invariance
Bilinear
Phase Response Comparison


Figure 1: Chebyshev Type-I LPF and HPF Responses
Chebyshev LPF Magnitude (Ord'''N = 6)
Chebyshev LPF Phase Resp'''se
Chebyshev HPF Magnitude (Ord'''N = 6)
Chebyshev HPF Phase Resp'''se
Magnitude (dB)
Phase (degrees)
Frequency (Hz)


Impulse Response of IIR Realizations
Direct Form-I
Direct Form-II
Cascade
Parallel
Amplitude
n


Numerical Error
DF-I vs DF-II
DF-I vs Cascade
DF-I vs Parallel
Absolute Error
n


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
ADCEXP1B.m
APP.M
exp1.m
exp1_1_ss.m
exp1_2_ss.m
exp1_dsp.m
exp2_1.m
exp2_3.m
exp2_dsp.m
exp3_1_ss.m
exp3_dsp.m
exp4_dsp.m
exp4ss_1.m
Workspace
Name
Value
Size
Class
a
[1,-2.3695,2.31...
1×5
double
b
[0.0048,0.0193...
1×5
double
D
[0;8.2420e-13;...
3×1
double
E
[0;1.9221e-15;...
3×1
double
g
0.0048
1×1
double
i
4
1×1
double
k
0.0256
1×1
double
n
1×50 double
1×50
double
N
50
1×1
double
p
[0.6604 + 0.44...
4×1
double (compl...
r
[-0.2698 + 0.1...
4×1
double (compl...
sos
2×6 double
2×6
double
Editor
exp1_dsp.m
exp3_dsp.m
exp2_dsp.m
exp4_dsp.m
exp5_dsp.m
/MATLAB Drive/exp5_dsp.m
clc; clear; close all;
%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];
N = 50;
x = [1 zeros(1,N-1)];
% Unit impulse
n = 0:N-1;
%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x);
% Direct Form-I
y2 = filter(b,a,x);
% Direct Form-II
%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);
yc = g*x;
for k = 1:size(sos,1)
yc = filter(sos(k,1:3),sos(k,4:6),yc);
end
Figures
Figure 1
Figure 2
Figure 3
Figure 4
Frequency Response of IIR Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
New to MATLAB? See resources for Getting Started.
Warning: Using only the real component of complex data.
> In matlab.graphics.chart.internal.getRealData (line 54)
In stem (line 63)
In exp5_dsp (line 62)
>>


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
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LABELS AND ANNOTATIONS
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exp3_1_ss.m
exp3_dsp.m
exp4_dsp.m
exp4ss_1.m
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1×5
double
D
[0;8.2420e-13;...
3×1
double
E
[0;1.9221e-15;...
3×1
double
g
0.0048
1×1
double
i
4
1×1
double
k
0.0256
1×1
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1×50 double
1×50
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N
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double (compl...
r
[-0.2698 + 0.1...
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Figure 3
Figure 4
Pole-Zero Plot
Imaginary Part
Real Part
Command Window
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>>